

Overview of the nervous system

The human nervous system is the most complex product of evolution. The activity of its billions of neurones and glia constitutes the physical basis of consciousness, experience, thought and behaviour. Research to elucidate the systems and circuits involved in normal neural development and function is one of the most active areas of contemporary biology and medicine, and yet our understanding of these processes, which is essential for translational success, remains far from complete.

The nervous system has two major divisions, the central nervous system (CNS) and the peripheral nervous system (PNS). The CNS consists of the brain, spinal cord, optic nerve and retina, and contains the majority of neuronal cell bodies. The PNS includes all nervous tissue outside the CNS and consists of the cranial and spinal nerves, the peripheral autonomic nervous system (ANS) and the special senses (taste, olfaction, vision, hearing and balance). It is composed mainly of the axons of sensory and motor neurones that pass between the CNS and the body. The ANS is subdivided into sympathetic and parasympathetic components. It consists of neurones that innervate secretory glands and cardiac and smooth muscle, and is concerned primarily with control of the internal environment. Neurones in the wall of the gastrointestinal tract form the enteric nervous system (ENS) and are capable of sustaining local reflex activity that is independent of the CNS. The ENS contains as many intrinsic neurones in its ganglia as the entire spinal cord and is often considered as a third division of the nervous system.

The detailed microstructure of the nervous system is described in Chapter 3 and the development of the nervous system is described in Chapter 17.

CENTRAL NERVOUS SYSTEM

The brain and spinal cord (see Fig. 45.1A) contain the great majority of neuronal cell bodies in the nervous system. In many parts of the CNS, neuronal cell bodies are grouped together and are, more or less, segregated from axons. The generic term for such collections of cell bodies is grey matter. Small aggregations of neuronal cell bodies, which usually share a common functional role, are termed nuclei. Neuronal dendrites and synaptic interactions are mostly confined to grey matter. Axons tend to be grouped together to form white matter, so called because axons are often ensheathed in myelin, which confers a paler colouration. Axons that pass between similar origins or destinations within the CNS tend to run together in defined pathways, or tracts. These often cross the midline (decussate), which means that one half of the body is, in many respects, controlled by, and sends information to, the opposite side of the brain.

Some groups of neurones in the spinal cord and brainstem that subserve similar functions are organized into longitudinal columns. The neurones in these columns may be concentrated into discrete, discontinuous nuclei in some areas, e.g. the cranial nerve nuclei of the brainstem, or they may form more or less continuous longitudinal bands, as in much of the spinal cord (Fig. 16.1). Efferent neurones constitute three such columns. The axons of motor neurones in the somatic motor column innervate muscles derived from head somites. The two other columns are related to specialized features of head morphology: neurones in the branchial motor column innervate muscles derived from the wall of the embryonic pharynx (branchial muscles) and the visceral motor column supplies preganglionic parasympathetic fibres to glands and visceral smooth muscle. Four longitudinal cell columns are related to sensory functions. The general somatic sensory column essentially deals with general sensory information from the head; special somatic sensory neurones are related to the special senses and receive vestibular and auditory input; general visceral sensory neurones deal with information from widespread and varied visceral

sensory endings; and special visceral sensory neurones are related to the special sense of taste.

The spinal cord and brain communicate with the rest of the body via the spinal and cranial nerves respectively. These nerves contain afferent fibres that bring information into the CNS from sensory receptors, and efferent fibres that convey instructions from the CNS to peripheral effector organs. The spinal cord and brainstem can control many aspects of bodily function by reflex action mediated via interconnections of varying complexity between the afferent and efferent components of the spinal and cranial nerves. Numerous descending connections from the brain may modulate this activity. Afferent input to the spinal cord and brainstem is channelled into various ascending pathways, some of which eventually impinge upon the cerebral cortex, conferring conscious awareness.

To provide the energy required by constant neuronal activity, the CNS has a high metabolic rate and a rich blood supply (Ch. 19). A blood–brain barrier controls the neuronal environment and imposes

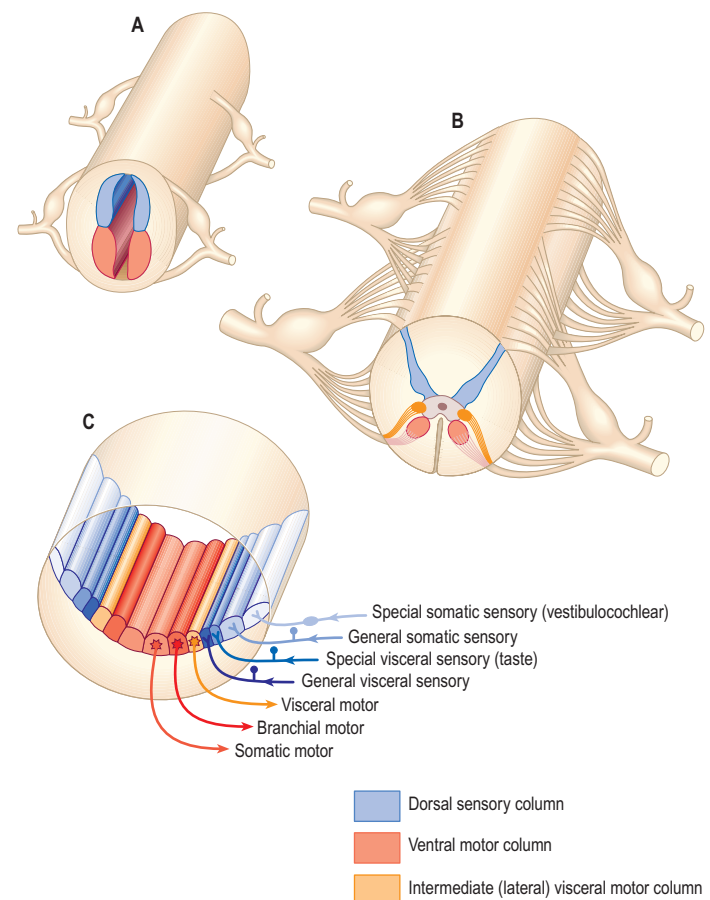


Fig. 16.1 The arrangement of sensory and motor cell columns in the spinal cord and brainstem. In each of parts A, B and C, dorsal is towards the top of the diagram. **A**, The organization of the primitive spinal cord with a dorsal sensory column, a ventral motor column, and segmentally arranged dorsal and ventral nerve roots. **B**, The adult thoracic spinal cord with sensory and somatic motor columns and an additional intermediate (lateral) visceral motor column. **C**, The arrangement of multiple longitudinal columns in the brainstem, where the motor column is now subdivided into three, and the sensory column into four. For further information about the embryological aspects of the early nervous system, consult Chapter 17.

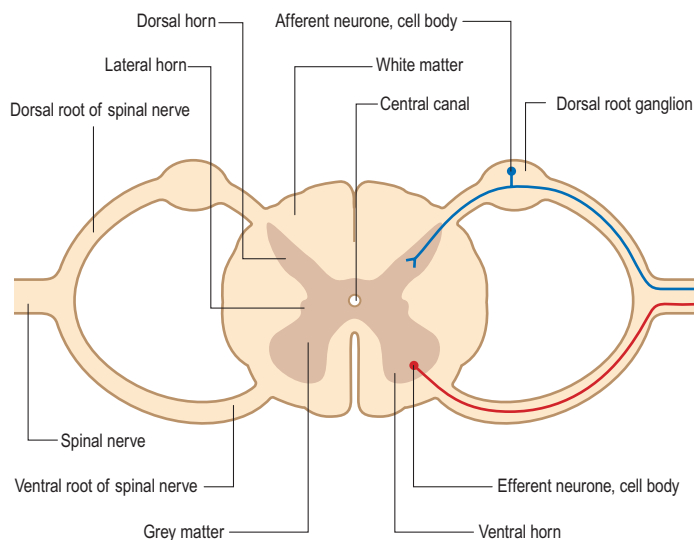


Fig. 16.2 A transverse section through the spinal cord, illustrating the disposition of grey and white matter and the attachment of dorsal and ventral spinal nerve roots.

severe restrictions on the types of substance that can pass from the bloodstream into nervous tissue.

SPINAL CORD

The spinal cord lies within the vertebral column, in the upper two-thirds of the vertebral canal, and is continuous rostrally with the medulla oblongata of the brainstem. For the most part, the spinal cord receives afferent input from, and controls the functions of, the trunk and limbs. Afferent and efferent connections between the periphery and the spinal cord travel in 31 pairs of segmentally arranged spinal nerves that attach to the cord as a linear series of dorsal and ventral rootlets. Adjacent groups of rootlets unite to form dorsal and ventral roots that combine to form the spinal nerves proper (Fig. 16.2). The dorsal and ventral roots are functionally distinct. Dorsal roots carry primary afferent nerve fibres from neuronal cell bodies located in dorsal root ganglia, whereas ventral roots carry efferent fibres from neuronal cell bodies located in the spinal grey matter.

Internally, the spinal cord consists of a central core of grey matter surrounded by white matter. The grey matter is configured in a characteristic H, or butterfly, shape that has projections known as dorsal (posterior) and ventral (anterior) horns (Fig. 16.3). In general, neurones situated in the dorsal horn are primarily concerned with sensory functions whilst those in the ventral horn are mostly associated with motor activities. At thoracic and upper lumbar levels of the spinal cord a small lateral horn is additionally present, marking the location of the cell bodies of preganglionic sympathetic neurones. The central canal, a vestigial component of the ventricular system, lies at the centre of the spinal grey matter and runs the length of the cord. The white matter of the spinal cord consists of ascending and descending tracts that link spinal cord segments to one another and the spinal cord to the brain.

BRAIN

The brain (encephalon) lies within the cranium. The brain receives information from, and controls the activities of, the trunk and limbs, mainly through connections with the spinal cord. It also possesses 12 pairs of cranial nerves through which it communicates mostly with structures of the head and neck. The brain is divided into major regions on the basis of ontogenetic growth and phylogenetic principles (Figs. 16.4–16.6). Ascending in sequence from the spinal cord, the principal divisions are the rhombencephalon or hindbrain, the mesencephalon or midbrain, and the prosencephalon or forebrain.

The rhombencephalon is subdivided into the myelencephalon or medulla oblongata, metencephalon or pons, and the cerebellum. The medulla oblongata, pons and midbrain are collectively referred to as the brainstem, which lies upon the basal portions of the occipital and sphenoid bones (clivus). The medulla oblongata is the most caudal part of the brainstem and is continuous with the spinal cord below the level of the foramen magnum. The pons lies rostral to the medulla and is distinguished by a mass of transverse nerve fibres that connect it to the cerebellum. The midbrain is a short segment of brainstem, rostral to

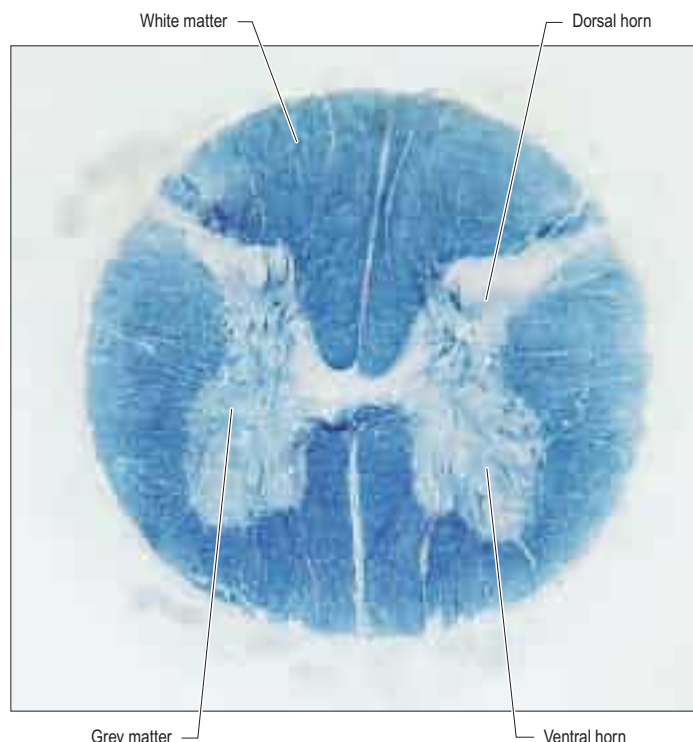


Fig. 16.3 A transverse section through the spinal cord at lumbar level. The section has been stained for nerve fibres, leaving the grey matter relatively unstained. (Figure enhanced by B Crossman.)

the pons. The cerebellum consists of paired hemispheres united by a median vermis and lies within the posterior cranial fossa, dorsal to the pons, medulla and caudal midbrain, with all of which it has numerous fibre connections.

The prosencephalon is subdivided into the diencephalon and the telencephalon. The diencephalon equates mostly to the thalamus and hypothalamus, but also includes the smaller epithalamus and subthalamus. The telencephalon is mainly composed of the two cerebral hemispheres or cerebrum. The diencephalon is almost completely embedded in the cerebrum and is, therefore, largely hidden from the exterior. The human cerebrum constitutes the major part of the brain. It occupies the anterior and middle cranial fossae and is directly related to the cranial vault. It consists of two cerebral hemispheres. The surface of each hemisphere is convoluted into a complex pattern of ridges (gyri) and furrows (sulci). Internally, each hemisphere has an outer layer of grey matter, the cerebral cortex, beneath which lies a thick mass of white matter (Fig. 16.7). One of the most important components of the cerebral white matter, the internal capsule (see Fig. 16.9), contains nerve fibres that pass to and from the cerebral cortex and lower levels of the neuraxis. Several large nuclei of grey matter, the basal ganglia or basal nuclei, are partly embedded in the subcortical white matter. Nerve fibre connections between corresponding areas on either side of the brain cross the midline within commissures, by far the largest being the corpus callosum.

During prenatal development, the walls of the neural tube thicken greatly but never completely obliterate the central lumen. Although the latter remains in the spinal cord as the vestigial central canal, it becomes greatly expanded in the brain to form a series of interconnected cavities or ventricles (Ch. 18). In two regions, the fore- and hindbrain, parts of the roof of the neural tube do not generate nerve cells but become thin, folded sheets of highly vascular secretory tissue, the choroid plexuses. These secrete cerebrospinal fluid that fills the ventricles. The cavity of the rhombencephalon becomes expanded to form the fourth ventricle, which lies dorsal to the pons and upper half of the medulla. Caudally, the fourth ventricle is continuous with a canal in the caudal medulla and, through this, with the central canal of the spinal cord. The fourth ventricle is continuous with the subarachnoid space through three openings in its roof (the foramina of Luschka and of Magendie). At its rostral extent, the fourth ventricle is continuous with a narrow channel, the cerebral aqueduct, which passes through the midbrain. The rostral end of the cerebral aqueduct opens out into the third ventricle, a narrow, slit-like, midline cavity that is bounded laterally by the diencephalon. At the rostral end of the third ventricle, a small aperture on each side leads into the large lateral ventricle that is located within each cerebral hemisphere (see Fig. 16.4C).

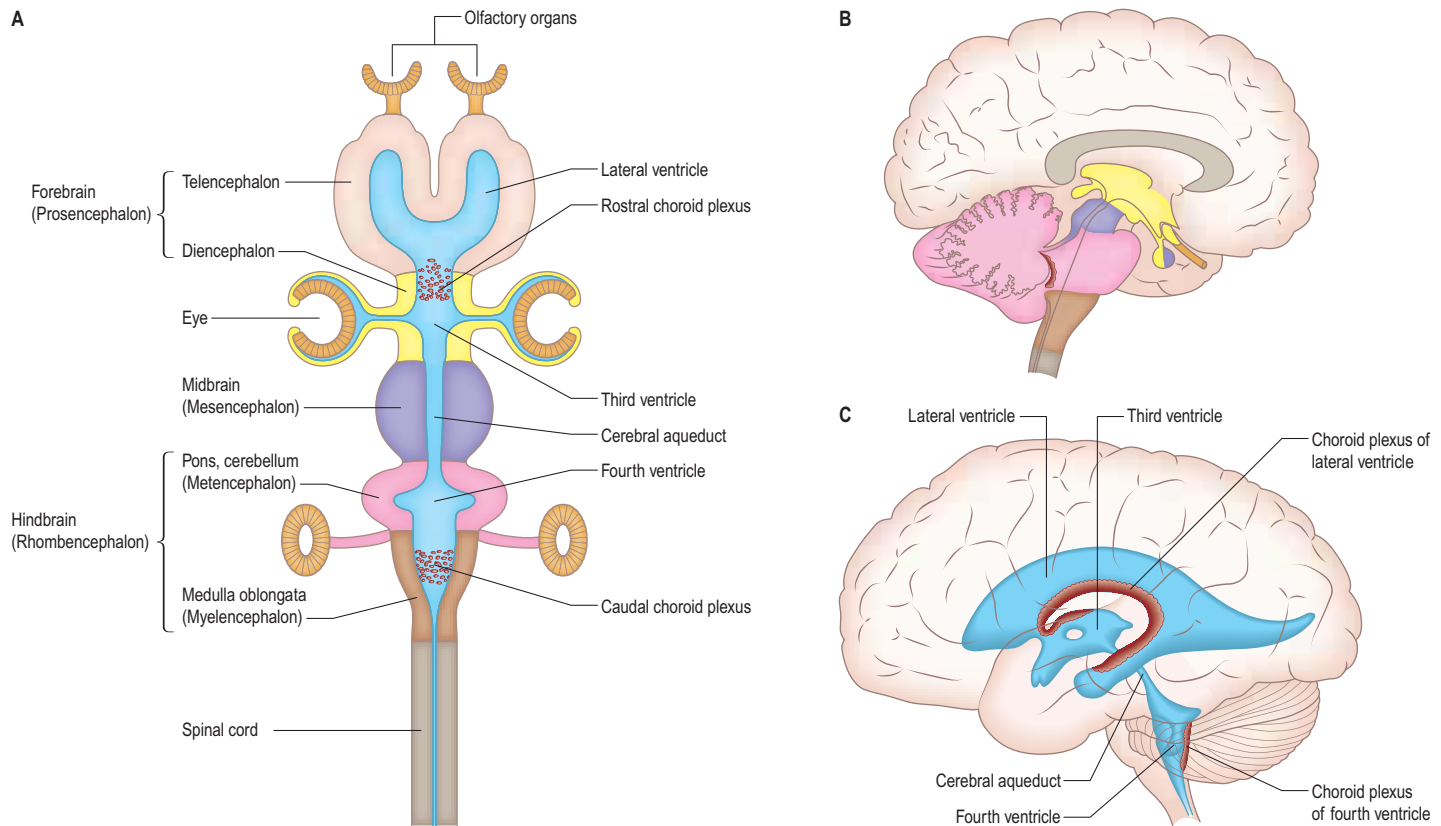


Fig. 16.4 Nomenclature and arrangement of the major divisions of the brain. **A**, The major features of the basic brain plan, including their relationships to the major special sensory organs of the head. **B**, The corresponding regions in the adult brain, seen in sagittal section. **C**, The organization of the ventricular system in the brain.

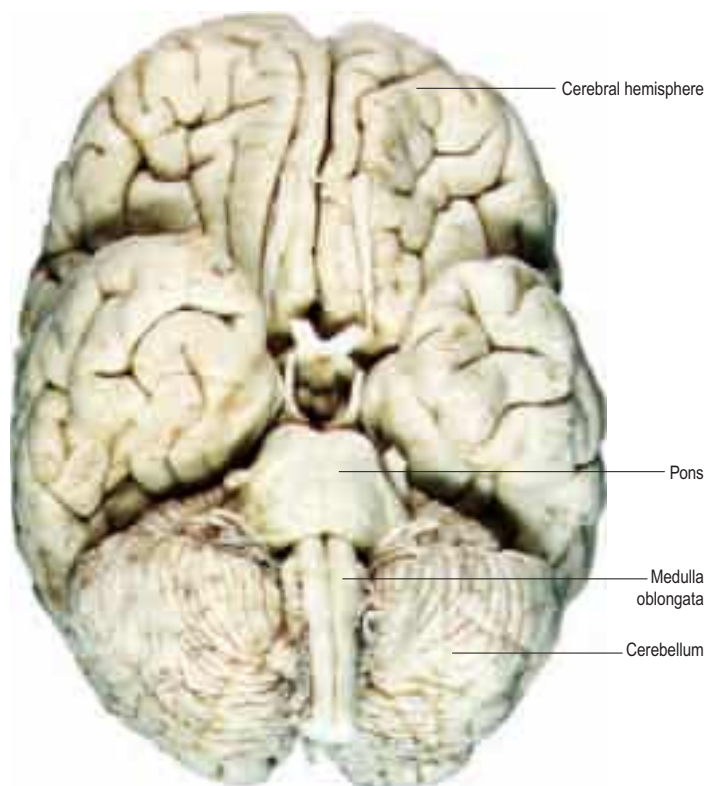


Fig. 16.5 The base of the brain, showing the major divisions and cranial nerves (see also Fig. 19.2). (Figure enhanced by B Crossman.)

OVERVIEW OF ASCENDING SENSORY PATHWAYS

Sensory modalities are conventionally described as being either special senses or general senses. The special senses are olfaction, vision, taste, hearing and vestibular function. Afferent information is encoded by highly specialized sense organs and transmitted to the brain in cranial nerves I, II, VII, VIII and IX.

The general senses include touch, pressure, vibration, pain, thermal sensation and proprioception (perception of posture and movement). Stimuli from the external and internal environments activate a diverse range of receptors in the skin, viscera, muscles, tendons and joints (Ch. 3). Afferent impulses from the trunk and limbs are conveyed to the spinal cord in spinal nerves, while those from the head are carried to the brain in cranial nerves. The detailed anatomy of the complex pathways by which the various general senses impinge on consciousness levels is better understood by reference to certain common organizational principles. Whilst undoubtedly oversimplified and subject to exceptions, this schema is helpful in emphasizing the essential similarities that exist between the ascending sensory systems.

In essence, ascending sensory projections related to the general senses consist of a sequence of three neurones that extends from peripheral receptor to contralateral cerebral cortex (Fig. 16.8). These are often referred to as primary, secondary and tertiary neurones or first-, second- and third-order neurones. Primary afferents have peripherally located sensory endings and cell bodies that lie in dorsal root ganglia or the sensory ganglia associated with certain cranial nerves. Their axons enter the CNS through spinal or cranial nerves and terminate by synapsing on the cell bodies of ipsilateral second-order neurones; the precise location of this termination depends on the modality.

Primary afferent fibres carrying pain, temperature and coarse touch/pressure information from the trunk and limbs terminate in the dorsal horn of the spinal grey matter, near their point of entry into the spinal cord. Homologous fibres from the head terminate in the trigeminal sensory nucleus of the brainstem. The cell bodies of second-order neurones are located in either the dorsal horn or the trigeminal sensory nucleus. Their axons decussate and ascend to the ventral posterior nucleus of the contralateral thalamus as the spinothalamic or the trigeminothalamic tract, respectively; they synapse on the cell bodies of third-order neurones in the thalamus. Axons of third-order neurones pass through the internal capsule to reach the cerebral cortex, terminating in the postcentral gyrus of the parietal lobe, which is also known as the primary somatosensory cortex.

Primary afferent fibres carrying proprioceptive information and fine (discriminative) touch from the trunk and limbs ascend ipsilaterally in the spinal cord as the dorsal columns (fasciculus gracilis and fasciculus cuneatus); they end by synapsing on second-order neurones in the dorsal column nuclei (nucleus gracilis and nucleus cuneatus) of the medulla. Axons of second-order neurones decussate in the medulla and then ascend as the medial lemniscus to the ventral posterior nucleus of



Fig. 16.6 A sagittal section of the brain. (Figure enhanced by B Crossman.)

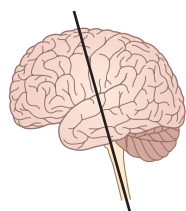
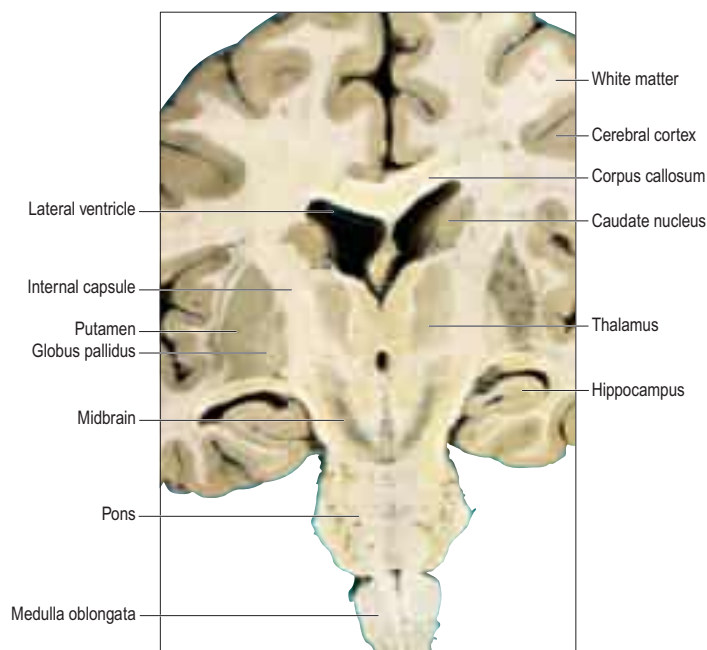


Fig. 16.7 A section through the cerebral hemisphere and brainstem, showing the disposition of grey and white matter, the basal ganglia and the internal capsule. (Dissection by EL Rees; photograph by Kevin Fitzpatrick, London.)

the contralateral thalamus, where they synapse on the cell bodies of third-order neurones. Axons of third-order neurones pass through the internal capsule to reach the cerebral cortex, terminating in the primary somatosensory cortex. A similar homologous projection exists for afferents derived from the head.

OVERVIEW OF DESCENDING MOTOR PATHWAYS

Corticofugal fibres descend through the internal capsule and pass into the brainstem, where many of them terminate, innervating the cranial nerve nuclei and other brainstem nuclei such as the red nucleus, reticular nuclei, olivary nuclei, etc. The term 'corticobulbar' has been used for many years to describe cortical projections to brainstem nuclei but it is a misnomer. The suffix 'bulbar' is derived from 'bulb', an obsolete name for the medulla oblongata: clearly, corticofugal fibres that terminate in the midbrain and/or pons cannot be described as corticobulbar. The term 'corticobulbar' has been replaced by 'corticonuclear'.

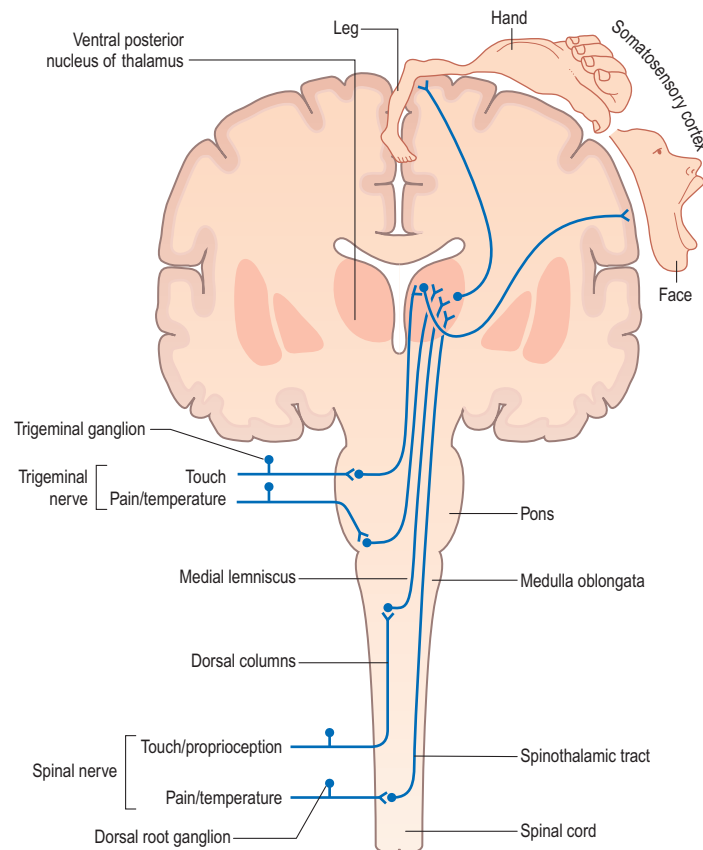


Fig. 16.8 The organization of general sensory pathways showing first-order, second-order and third-order neurones.

Terminologia Anatomica specifies cortical axons that innervate motor or sensory nuclei of cranial nerves according to their connections as follows: *fibrae corticonucleares bulbi* (medullary corticonuclear fibres); *fibrae corticonuclearis pontinis* (pontine corticonuclear fibres); and *fibrae corticonuclearis mesencephali* (mesencephalic corticonuclear fibres).

Corticospinal (pyramidal tract) fibres originate from widespread regions of the cerebral cortex, including the primary motor cortex of the frontal lobe where the opposite half of the body is represented in a detailed somatotopic fashion. The fibres descend throughout the length of the brainstem. The majority then cross to the contralateral side in the motor decussation of the pyramids in the medulla. Thereafter, they continue caudally as the lateral corticospinal tract of the spinal cord, which terminates in association with interneurons and motor neurones of the spinal grey matter (Fig. 16.9). The principal function of the corticonuclear and corticospinal tracts is the control of fine, fractionated movements, particularly of those parts of the body where delicate muscular control is required. These tracts are particularly important in speech (corticonuclear tract) and movements of the hands (corticospinal tract).

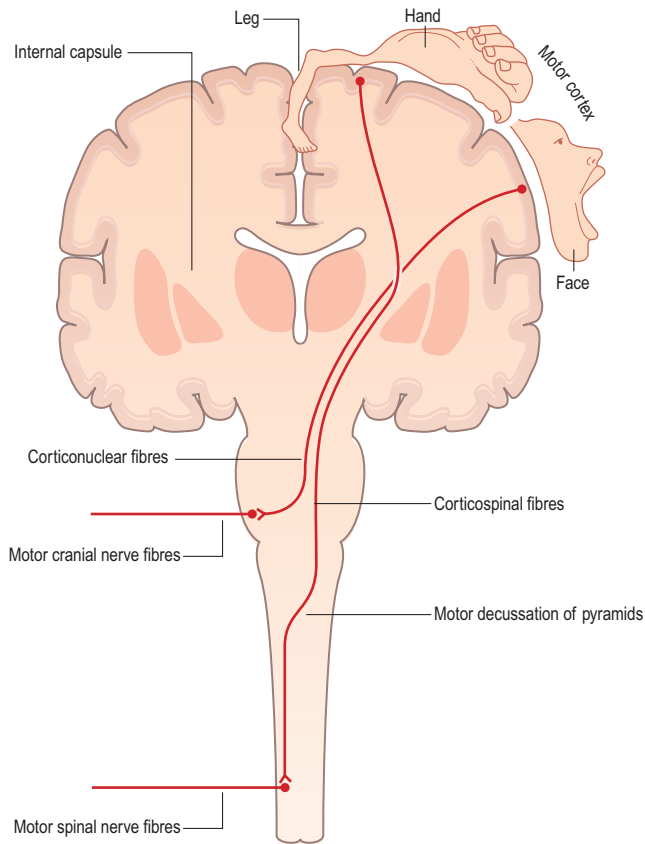


Fig. 16.9 The corticospinal and corticonuclear tracts.

The concept of 'upper' and 'lower' motor neurones is fundamental in clinical neurology because the motor signs and symptoms of damage to each category are different and are indicative of the anatomical site of the lesion. Lower motor neurones are the alpha motor neurones located in the brainstem and spinal cord that innervate the extrafusal muscle fibres of skeletal muscle. The term upper motor neurones refers collectively to all the descending pathways that impinge upon the activity of lower motor neurones but, in common parlance, the term is often equated with the corticonuclear and corticospinal tracts. The terms upper and lower motor neurone lesion are used clinically to distinguish, for example, between the effects of a stroke in the internal capsule (a typical upper motor neurone lesion) and those of motor neurone disease (a typical lower motor neurone lesion).

Lower motor neurone lesions cause paralysis or paresis of specific muscles because they have lost their direct innervation. There is also loss or reduction of tendon reflex activity and reduced muscle tone. Spontaneous muscular contractions (fasciculation) occur and affected muscles atrophy over time. Upper motor neurone lesions cause paralysis or paresis of movements as a result of loss of higher control. There is increased tendon reflex activity and increased muscle tone, and no muscle atrophy occurs. A positive plantar (Babinski) reflex is present. The combination of paralysis, increased tendon reflex activity and hypertonia is referred to as spasticity.

The pathophysiology underlying the symptoms of upper motor neurone lesions is complex. This is because many descending pathways other than the corticonuclear and corticospinal tracts exist and they also influence lower motor neurone activity. These pathways include corticofugal projections to the brainstem (e.g. corticoreticular and corticopontine) that traverse the internal capsule and numerous pathways that originate within the brainstem itself (e.g. reticulospinal, vestibulospinal). Clearly, these pathways may be compromised to varying extents, depending on the site of a lesion. Their involvement is believed to be important in the pathophysiological mechanisms that underlie the generation of spasticity. Pure corticospinal tract lesions, which are exceedingly rare in humans because corticospinal tract fibres lie in close relationship to other pathways throughout most of their course, are believed specifically to cause deficits in delicate, fractionated movements and to induce the positive plantar reflex.

Two other major systems that contribute to the control of movement are the basal ganglia and the cerebellum. The basal ganglia are a group of large subcortical nuclei, the major components being the caudate nucleus, putamen and globus pallidus (see Fig. 16.7; Ch. 24). These structures have important connections with the cerebral cortex, certain

nuclei of the thalamus and subthalamus, and with the brainstem. They appear to be involved in the selection of appropriate behavioural patterns/movements and the suppression of inappropriate ones. Disorders of the basal ganglia cause either too little movement (akinesia) or abnormal involuntary movements (dyskinesias), as well as tremor and abnormalities of muscle tone (Crossman and Neary 2014). The basal ganglia are sometimes described as being part of the so-called 'extrapyramidal (motor) system'. This term is used to distinguish between the effects of basal ganglia disease and those of damage to the 'pyramidal' (corticospinal) system. However, the progressive elucidation of the anatomy of the basal ganglia and of the pathophysiology of motor disorders has revealed the close functional interrelationship between the two 'systems', and has rendered the terms that distinguish them largely obsolete (Brodal 1981). The cerebellum (Ch. 22) has rich connections with the brainstem, particularly the reticular and vestibular nuclei, and with the thalamus. It is concerned with the coordination of movement; cerebellar disorders cause ataxia, intention tremor and hypotonia.

PERIPHERAL NERVOUS SYSTEM

The PNS is composed mainly of spinal nerves, cranial nerves, their ganglia and their ramifications that carry afferent and efferent neurones between the CNS and the rest of the body. It also includes the peripheral part of the autonomic nervous system, notably the sympathetic trunks and ganglia, and the enteric nervous system, composed of plexuses of nerve fibres and cell bodies in the wall of the gastrointestinal tract.

SPINAL NERVES

Spinal nerves are the means by which the CNS receives information from, and controls the activities of, the trunk and limbs. Spinal nerves are considered in detail elsewhere on a regional basis (Sections 4–9).

There are 31 pairs of spinal nerves (8 cervical, 12 thoracic, 5 lumbar, 5 sacral, 1 coccygeal) and these contain a mixture of sensory and motor fibres. They originate from the spinal cord as continuous series of dorsal and ventral nerve rootlets. Adjacent groups of rootlets fuse to form dorsal and ventral roots, which then merge to form the spinal nerves proper. The dorsal roots of spinal nerves contain afferent nerve fibres from cell bodies located in dorsal root ganglia. These cells give off both centrally and peripherally directed processes and do not have synapses on their cell bodies. The ventral roots of spinal nerves contain efferent fibres from cell bodies located in the spinal grey matter. These include motor neurones innervating skeletal muscle and preganglionic autonomic neurones.

Spinal nerves exit from the vertebral canal via their corresponding intervertebral foramina. They then divide to form a large ventral (anterior) ramus and a smaller dorsal (posterior) ramus. In general terms, the ventral ramus innervates the limbs together with the muscles and skin of the anterior part of the trunk. The dorsal ramus innervates the post-vertebral muscles and the skin of the back. The nerve fibres within the ventral rami serving the upper and lower limbs are redistributed within brachial and lumbosacral plexuses, respectively.

CRANIAL NERVES

Cranial nerves are the means by which the brain receives information from, and controls the activities of, the head and neck, and to a lesser extent the thoracic and abdominal viscera. The component fibres, their route of exit from the cranial cavity, their subsequent peripheral course and the distribution and functions of the cranial nerves are considered in detail elsewhere on a regional basis (Sections 4, 7, 8). Their origins, destinations and connections within the CNS are considered in this section.

There are 12 pairs of cranial nerves that are individually named and numbered (Roman numerals) in a rostrocaudal sequence (Table 16.1). Unlike spinal nerves, only some cranial nerves are mixed in function, carrying both sensory and motor fibres; others are purely sensory or purely motor. The first cranial nerve (I; olfactory) has an ancient lineage and is derived from the forerunner of the cerebral hemisphere. It retains this unique position through the connections of the olfactory bulb, and is the only sensory cranial nerve that projects directly to the cerebral cortex rather than indirectly via the thalamus. The areas of cerebral cortex receiving olfactory input have a primitive cellular organization and are an integral part of the limbic system, which is concerned with the emotional aspects of behaviour. The second cranial nerve (II; optic)

OVERVIEW OF THE NERVOUS SYSTEM

Table 16.1 Summary of cranial nerves

No.	Name	Components	Cells of origin	Main central connections	Exit from skull	Distribution and function
I	Olfactory	Special somatic afferent	Olfactory receptor cells in nasal mucosa	To olfactory bulb, anterior perforated substance, piriform cortex	Foramina of cribriform plate	Olfaction
II	Optic	Special somatic afferent	Ganglion cells of retina	To lateral geniculate nucleus of thalamus, pretectal area of midbrain	Optic canal	Vision
III	Oculomotor	Somatic efferent	Oculomotor nucleus	From visual cortical areas, medial longitudinal fasciculus	Superior orbital fissure	Motor to medial rectus, inferior rectus, superior rectus, inferior oblique, levator palpebrae superioris
III	Oculomotor	General visceral efferent	Edinger–Westphal nucleus	From pretectal area of midbrain	Superior orbital fissure	Parasympathetic to sphincter pupillae and ciliary muscles, via ciliary ganglion, mediating pupillary constriction and accommodation
IV	Trochlear	Somatic efferent	Trochlear nucleus	From visual cortical areas, medial longitudinal fasciculus	Superior orbital fissure	Motor to superior oblique
Trigeminal division:						
V	Ophthalmic (V ₁)	General somatic afferent	Trigeminal (semilunar) ganglion	To trigeminal sensory nucleus	Superior orbital fissure	Sensation from forehead, scalp, eyelids, nose and eye, including conjunctiva
	Maxillary (V ₂)	General somatic afferent	Trigeminal (semilunar) ganglion	To trigeminal sensory nucleus	Foramen rotundum	Sensation from mid-face, lower eyelid, nasal cavity and paranasal sinuses, upper lip, maxillary teeth and palate
	Mandibular (V ₃)	General somatic afferent	Trigeminal (semilunar) ganglion	To trigeminal sensory nucleus	Foramen ovale	Sensation from lower face and scalp, tongue and floor of mouth, mandibular teeth and part of external ear
		General somatic afferent	Mesencephalic part of trigeminal sensory nucleus	To trigeminal sensory nucleus, cerebellum	Foramen ovale	Proprioception from muscles of mastication, temporomandibular joint
	Mandibular (V ₃)	Branchial (special visceral) efferent	Trigeminal motor nucleus	From precentral gyrus	Foramen ovale	Motor to muscles of mastication (temporalis, masseter, medial and lateral pterygoids), tensor tympani
VI	Abducens	Somatic efferent	Abducens nucleus	From visual cortical areas, medial longitudinal fasciculus	Superior orbital fissure	Motor to lateral rectus
VII	Facial	General somatic afferent	Geniculate ganglion	To trigeminal sensory nucleus, cerebellum	Internal acoustic meatus	Sensation from part of tympanic membrane, external acoustic meatus and skin behind pinna. Proprioception from facial muscles
VII	Facial	Special visceral afferent	Geniculate ganglion	To nucleus solitarius	Internal acoustic meatus	Taste from anterior two-thirds of tongue
VII	Facial	General visceral efferent	Superior salivatory nucleus	From olfactory areas, nucleus solitarius	Internal acoustic meatus	Parasympathetic to lacrimal gland and mucous membrane of nasal and oral cavities (via pterygopalatine ganglion) and submandibular and sublingual salivary glands (via submandibular ganglion) causing secretion and vasodilation
VII	Facial	Branchial (special visceral) efferent	Facial nucleus	From precentral gyrus	Internal acoustic meatus	Motor to 'muscles of facial expression', stapedius, posterior belly of digastric and stylohyoid
VIII	Vestibulocochlear					
	Vestibular division	Special somatic afferent	Vestibular ganglion	To vestibular nuclei, cerebellum	Internal acoustic meatus	Sensations of equilibrium and motion
	Cochlear division	Special somatic afferent	Spiral ganglion	To cochlear nuclei	Internal acoustic meatus	Hearing
IX	Glossopharyngeal	General somatic afferent	Glossopharyngeal ganglia	To trigeminal sensory nucleus	Jugular foramen	General sensation from posterior one-third of tongue, oropharynx and middle ear
IX	Glossopharyngeal	General visceral afferent	Glossopharyngeal ganglia	To nucleus solitarius	Jugular foramen	Sensory from carotid body (chemoreceptors) and carotid sinus (baroreceptors)
IX	Glossopharyngeal	Special visceral afferent	Glossopharyngeal ganglia	To nucleus solitarius	Jugular foramen	Taste from posterior one-third of tongue
IX	Glossopharyngeal	General visceral efferent	Inferior salivatory nucleus	From olfactory areas, nucleus solitarius	Jugular foramen	Parasympathetic to parotid salivary gland (via otic ganglion), causing secretion and vasodilation
IX	Glossopharyngeal	Branchial (special visceral) efferent	Nucleus ambiguus	From precentral gyrus	Jugular foramen	Motor to stylopharyngeus
X	Vagus	General somatic afferent	Superior (jugular) ganglion	To trigeminal sensory nucleus	Jugular foramen	General sensation from pharynx, larynx, trachea, oesophagus, part of auricle and external auditory meatus
X	Vagus	General visceral afferent	Inferior (nodose) ganglion	To nucleus solitarius	Jugular foramen	Sensory from thoracic and abdominal viscera
X	Vagus	General visceral efferent	Dorsal motor nucleus of vagus	From hypothalamus, nucleus solitarius	Jugular foramen	Parasympathetic to glands and smooth muscle in the pharynx, larynx, thoracic and abdominal viscera
X	Vagus	Branchial (special visceral) efferent	Nucleus ambiguus	From precentral gyrus, sensory nuclei of brainstem	Jugular foramen	Motor to pharyngeal, external laryngeal and oesophageal striated muscles
XI	*Accessory					
	Cranial root	Branchial (special visceral) efferent	Nucleus ambiguus	From precentral gyrus, sensory nuclei of brainstem	Jugular foramen	Motor to muscles of soft palate and intrinsic muscles of larynx (distributed via vagus)
	Spinal root	Branchial (special visceral) efferent	Spinal cord segments C1–C5	From precentral gyrus	Jugular foramen	Motor to sternocleidomastoid and trapezius
XII	Hypoglossal	Somatic efferent	Hypoglossal nucleus	From precentral gyrus	Hypoglossal canal	Motor to all intrinsic and extrinsic muscles of the tongue, except palatoglossus

Key

	General somatic afferent
	Special somatic afferent
	General visceral afferent
	Special visceral afferent
	Somatic efferent
	General visceral efferent
	Branchial (special visceral) efferent

*The existence of a 'cranial root' of the accessory nerve, with its cell bodies in the nucleus ambiguus and a distribution as given in Table 16.1, is controversial. There is a view that the 'spinal root' of the accessory nerve should be regarded as the accessory nerve proper (sometimes called the 'spinal accessory nerve') and that the vagus nerve supplies the muscles of the soft palate and the intrinsic muscles of the larynx. See Chapter 21 for further details.

consists of the axons of second-order visual neurones and these terminate in the thalamus. The other ten pairs of cranial nerves attach to the brainstem and most of their component fibres originate from, or terminate in, the cranial nerve nuclei of the brainstem.

The sensory fibres in individual spinal and cranial nerves have characteristic, but often overlapping, peripheral distributions. As far as the innervation of the body surface is concerned, the area that is supplied by a particular spinal or cranial nerve is referred to as a dermatome (Greenberg 2003), whereas the areas of skin supplied by particular peripheral nerves are often referred to as the peripheral nerve fields (or cutaneous nerve distributions). There is marked overlap between dermatomes of adjacent spinal nerves, particularly for the segments least affected by development of the limbs, i.e. the second thoracic to the first lumbar (Ladak et al 2013). In some regions, e.g. the upper anterior thoracic wall, the cutaneous nerves that supply adjoining areas are not derived from consecutive spinal nerves and in these instances the overlap between dermatomes is minimal; these boundaries give more reliable and clinically useful borders (Apok et al 2011). Maps of dermatome distribution are useful in clinical neurology as a guide to identify the location of pathology in patients with peripheral sensory deficits, but it is important to remember that they are approximations. Dermatome maps are somewhat inconsistently reported by different authors, reflecting the fact that the maps are composites that have been compiled to a large extent by clinical observations on patients with cranial or spinal nerve pathology, and also that considerable normal individual variation exists (Fig. 16.10) (Lee et al 2008). Common to all published maps are areas of sensory overlap along borders of contiguous spinal nerves and axial lines where non-contiguous spinal nerves meet, and across which very little or no sensory overlap occurs. Detailed dermatome maps are described on a regional basis. The motor axons of individual spinal and cranial nerves tend to innervate anatomically and functionally related groups of skeletal muscles, which are referred to as myotomes.

AUTONOMIC NERVOUS SYSTEM

The autonomic nervous system represents the visceral component of the nervous system. It consists of neurones located within both the CNS and the PNS that are concerned with the control of the internal environment, through the innervation of secretory glands and cardiac and smooth muscle. Its functions are, however, closely integrated with those of the somatic nervous system.

Visceral efferent pathways differ from their somatic equivalents in that the former are interrupted by peripheral synapses, there being a sequence of at least two neurones between the CNS and the target structure (Fig. 16.11). These are referred to as preganglionic and postganglionic neurones, respectively. The somata of preganglionic neurones are located in the visceral efferent nuclei of the brainstem and in the lateral grey columns of the spinal cord. Their axons, which are usually finely myelinated, exit from the CNS in certain cranial and spinal nerves, and then pass to peripheral ganglia, where they synapse with postganglionic neurones. The axons of postganglionic neurones are usually unmyelinated. Postganglionic neurones are more numerous than preganglionic ones; one preganglionic neurone may synapse with 15–20 postganglionic neurones, which permits the wide distribution of many autonomic effects.

The autonomic nervous system is divided into three major parts: sympathetic, parasympathetic and enteric. These differ in organization and structure but are functionally integrated. Most, but not all, structures innervated by the autonomic nervous system receive both sympathetic and parasympathetic fibres, whereas the enteric nervous system is a network of neurones intrinsic to the wall of the gastrointestinal tract.

Two long-held dogmas are that the sympathetic and parasympathetic nervous systems are functionally antagonistic (because activation of their respective efferents has opposing actions on target structures)

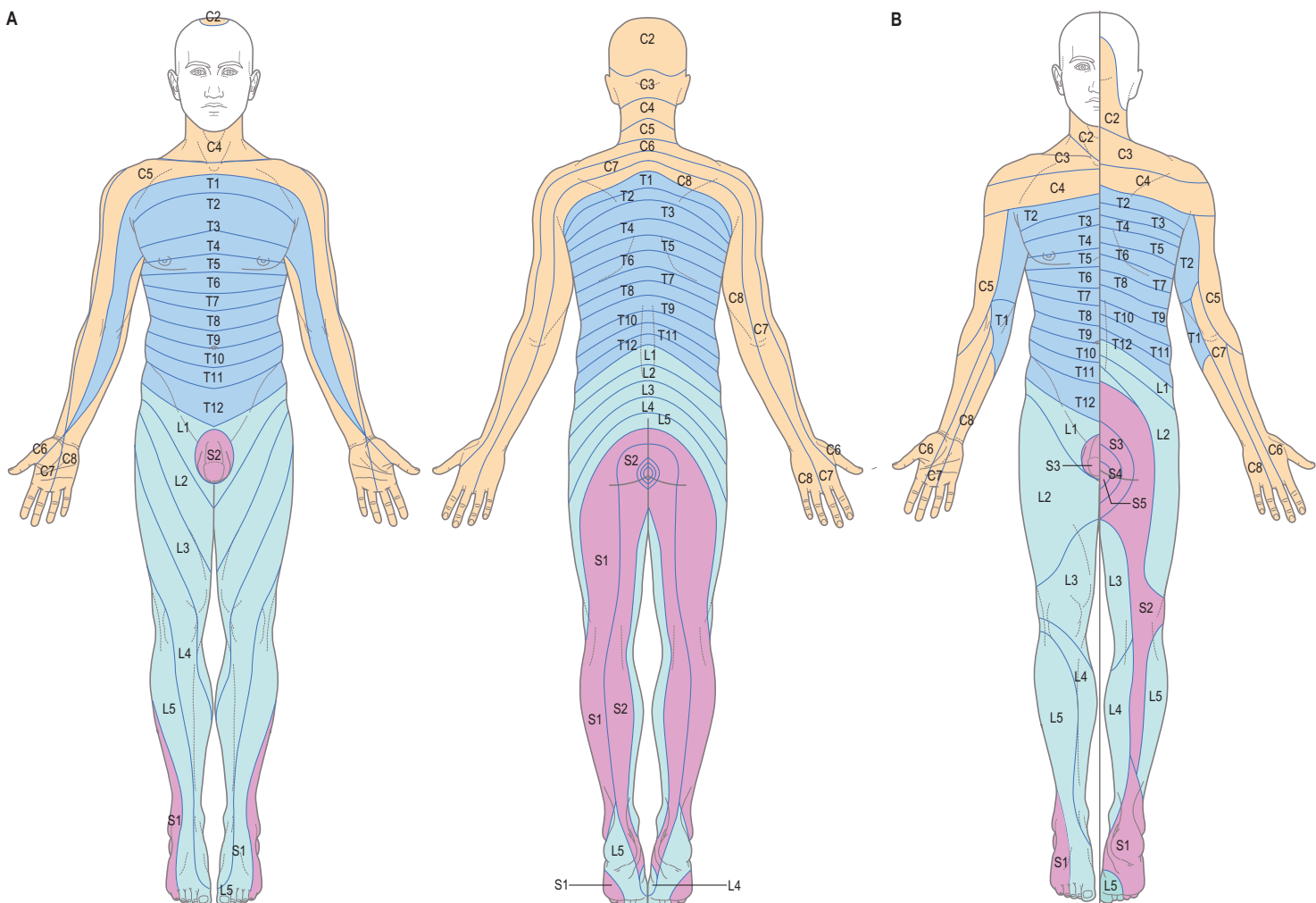


Fig. 16.10 Typical dermatome maps, illustrating some of the variations reported by different authors. **A**, Dermatome map based on Keegan JJ, Garrett FD 1948 The segmental distribution of the cutaneous nerves in the limbs of man. *Anat Rec* 102:409–439. Mainly used in literature in the USA.

B, Dermatome map based on Moffat DB 1993 Lecture Notes on Anatomy, 2nd edn. Oxford: Blackwell Scientific. Mainly used in literature in the UK.

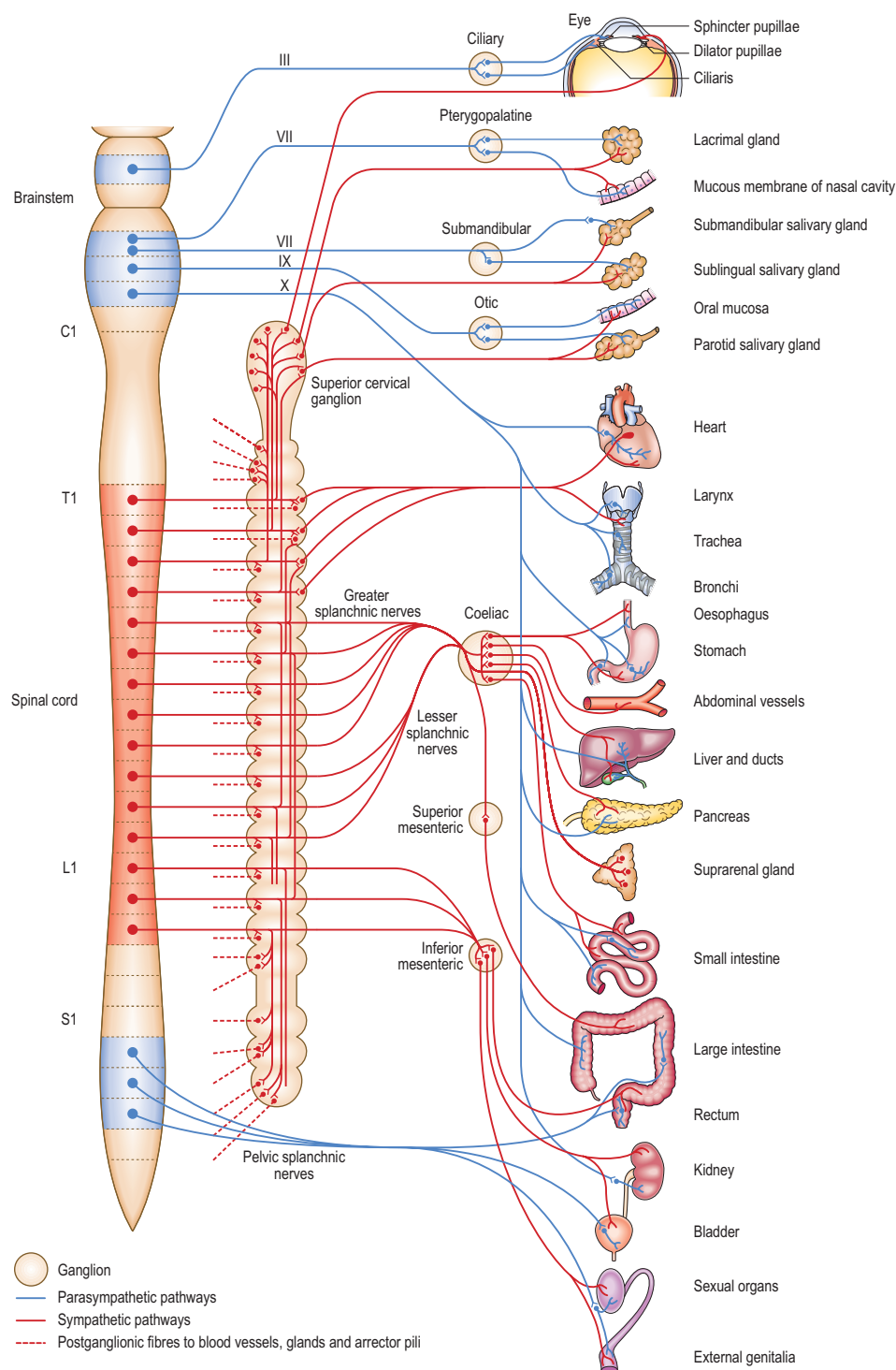


Fig. 16.11 Efferent pathways of the autonomic nervous system.

and that sympathetic reactions are mass responses whereas parasympathetic reactions are usually more localized. A better concept is that the two sets of neurones constitute an integrated system for the coordinated neural regulation of visceral and homeostatic functions. Moreover, even though widespread activation of the sympathetic nervous system may occur, e.g. in association with fear or rage, it is recognized that the sympathetic nervous system is also capable of discrete activation. In general, sympathetic activity results in the constriction of cutaneous arteries (increasing blood supply to the heart, muscles and brain), cardiac acceleration, an increase in blood pressure, contraction of sphincters and depression of peristalsis, all of which mobilize body energy stores for dealing with increased activity. Parasympathetic activity results in cardiac slowing and an increase in intestinal glandular and peristaltic activities, which may be considered to conserve body energy stores.

Autonomic activity is not initiated or controlled solely by the reflex connections of general visceral afferent pathways; nor do impulses in these pathways necessarily activate general visceral efferents. For example, in many situations demanding general sympathetic activity, the initiator is somatic and typically arises from the special senses or the skin. Rises in blood pressure and pupillary dilation may result

from the stimulation of somatic receptors in the skin and other tissues. Peripheral autonomic activity is integrated at higher levels in the brainstem and cerebrum, including various nuclei of the brainstem reticular formation, thalamus and hypothalamus, the limbic lobe and prefrontal neocortex, together with the ascending and descending pathways that interconnect these regions.

The classical concept of autonomic neurotransmission is that pre-ganglionic neurones of both sympathetic and parasympathetic systems are cholinergic and that postganglionic parasympathetic neurones are also cholinergic while those of the sympathetic nervous system are noradrenergic. The discovery of neurones that do not use either acetylcholine or noradrenaline (norepinephrine) as their primary transmitter, and the recognition of a multiplicity of substances in autonomic nerves which fulfil the criteria for a neurotransmitter or neuromodulator, have greatly complicated neuropharmacological concepts of the autonomic nervous system (Burnstock 2013). Thus, adenosine 5'-triphosphate (ATP), numerous peptides and nitric oxide have all been implicated in the mechanisms of cell signalling in the autonomic nervous system. The principal co-transmitters in sympathetic nerves are ATP and neuropeptide Y; in parasympathetic nerves vasoactive intestinal polypeptide (VIP); and in enteric nerves ATP, VIP and substance P.

SYMPATHETIC NERVOUS SYSTEM

The sympathetic trunks are two ganglionated nerve cords that extend on either side of the vertebral column from the cranial base to the coccyx. The ganglia are joined to spinal nerves by short connecting nerves called white and grey rami communicantes. Preganglionic axons join the trunk through the white rami communicantes while postganglionic axons leave the trunk in the grey rami. In the neck, each sympathetic trunk lies posterior to the carotid sheath and anterior to the transverse processes of the cervical vertebrae. In the thorax, the trunks are anterior to the heads of the ribs, in the abdomen they lie anterolateral to the bodies of the lumbar vertebrae, and in the pelvis they are anterior to the sacrum and medial to the anterior sacral foramina. Anterior to the coccyx the two trunks meet in a single, median, terminal ganglion. Cervical sympathetic ganglia are usually reduced to three by fusion. The internal carotid nerve, a continuation of the sympathetic trunk, issues from the cranial pole of the superior ganglion and accompanies the internal carotid artery through its canal into the cranial cavity. There are between 10 and 12 (usually 11) thoracic ganglia, 4 lumbar ganglia, and 4 or 5 ganglia in the sacral region.

The cell bodies of preganglionic sympathetic neurones are located in the lateral horn of the spinal grey matter of all thoracic segments and the upper two or three lumbar segments (Fig. 16.12). Their axons are myelinated with diameters of 1.5–4 μm . They leave the cord in the corresponding ventral nerve roots and pass into the spinal nerves, but soon leave in white rami communicantes to join the sympathetic trunk (Fig. 16.13). Neurones like those in the lateral grey column exist at other levels of the cord above and below the thoracolumbar outflow and small numbers of their fibres leave in other ventral roots.

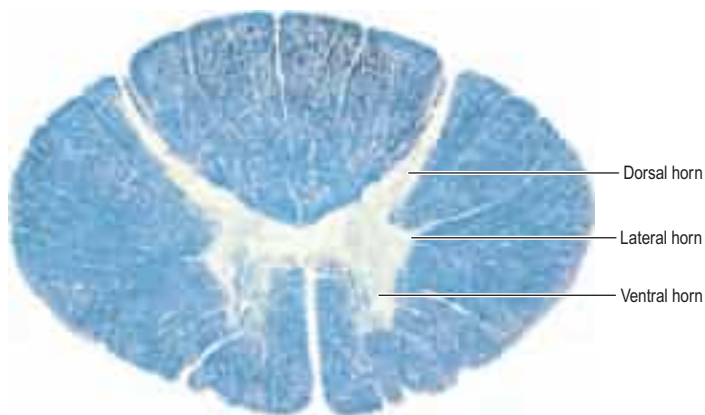


Fig. 16.12 A transverse section through the thoracic spinal cord. Preganglionic sympathetic neurones are located in the lateral horn. (Figure enhanced by B Crossman.)

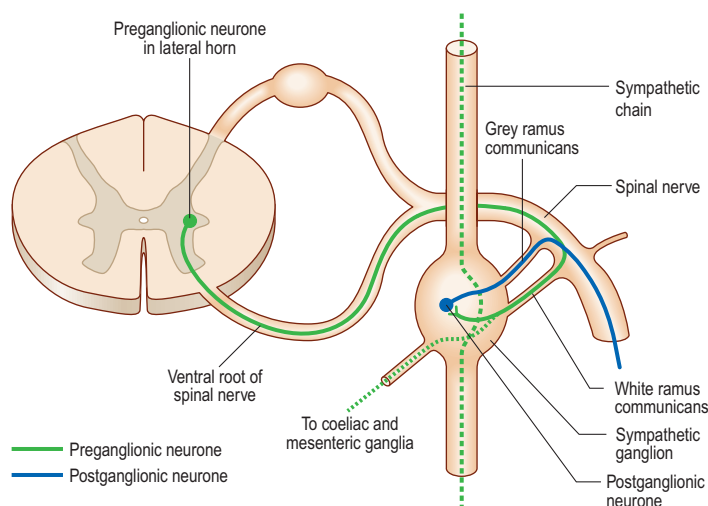


Fig. 16.13 Outflow from preganglionic sympathetic neurones in the lateral horn of the spinal cord. Preganglionic axons may synapse on a postganglionic neurone in a sympathetic ganglion at the same segmental level, or may pass into the sympathetic chain to synapse on a postganglionic neurone in a more rostral or caudal sympathetic ganglion, or may pass to a ganglion in an autonomic plexus in the abdominal cavity, e.g. the coeliac ganglion.

Preganglionic sympathetic neurones release acetylcholine as their principal neurotransmitter.

On reaching the sympathetic trunk, preganglionic fibres behave in one of several ways (see Fig. 16.13). They may synapse with neurones in the nearest ganglion, or traverse the nearest ganglion and ascend or descend in the sympathetic chain to end in another ganglion. A preganglionic fibre may terminate in a single ganglion or, through collateral branches, synapse with neurones in several ganglia. Preganglionic fibres may traverse the nearest ganglion, ascend or descend and, without synapsing, emerge in one of the medially directed branches of the sympathetic trunk to synapse in the ganglia of autonomic plexuses (mainly situated in the midline, e.g. around the coeliac and mesenteric arteries). More than one preganglionic fibre may synapse with a single postganglionic neurone. Uniquely, the suprarenal gland is innervated directly by preganglionic sympathetic neurones that traverse the sympathetic trunk and coeliac ganglion without synapse.

The somata of sympathetic postganglionic neurones are located mostly either in the ganglia of the sympathetic trunk or in ganglia in more peripheral plexuses. The axons of postganglionic neurones are, therefore, generally longer than those of preganglionic neurones, an exception being some of those that innervate pelvic viscera. The axons of ganglionic cells are unmyelinated. They are distributed to target organs in various ways. Those from a ganglion of the sympathetic trunk may return to the spinal nerve of preganglionic origin through a grey ramus communicans, which usually joins the nerve just proximal to the white ramus, and are then distributed through ventral and dorsal spinal rami to blood vessels, sweat glands, hairs, etc., in their zone of supply. Segmental areas vary in extent and overlap considerably. The extent of innervation of different effector systems, e.g. vasomotor, sudomotor, etc., by a particular nerve may not be the same. Alternatively, postganglionic fibres may pass in a medial branch of a ganglion direct to particular viscera, or innervate adjacent blood vessels, or pass along them externally to their peripheral distribution. They may ascend or descend before leaving the sympathetic trunk as described above. Many fibres are distributed along arteries and ducts as plexuses to distant effectors.

The principal neurotransmitter released by postganglionic sympathetic neurones is noradrenaline (norepinephrine). The sympathetic system has a much wider distribution than the parasympathetic. It innervates all sweat glands, the arrector pili muscles, the muscular walls of many blood vessels, the heart, lungs and respiratory tree, the abdominal/pelvic viscera, the oesophagus, the muscles of the iris, and the non-striated muscle of the urogenital tract, eyelids and elsewhere.

Postganglionic sympathetic fibres that return to the spinal nerves are vasoconstrictor to blood vessels, secretomotor to sweat glands and motor to the arrector pili muscles within their dermatomes. Those that accompany the motor nerves to voluntary muscles are probably only dilatory. Most, if not all, peripheral nerves contain postganglionic sympathetic fibres. Those reaching the viscera are concerned with general vasoconstriction, bronchial and bronchiolar dilation, modification of glandular secretion, pupillary dilation, inhibition of gastrointestinal muscle contraction, etc. A single preganglionic fibre probably synapses with the postganglionic neurones in only one effector system, which means that effects such as sudomotor and vasomotor actions can be separate.

PARASYMPATHETIC NERVOUS SYSTEM

Preganglionic parasympathetic neuronal cell bodies are located in certain cranial nerve nuclei of the brainstem (see Fig. 21.1) and in the intermediate grey matter of the second to fourth sacral segments of the spinal cord. Efferent fibres, which are myelinated, emerge from the CNS only in the oculomotor, facial, glossopharyngeal and vagus nerves, and in the second to fourth sacral spinal nerves. Preganglionic parasympathetic neurones are cholinergic.

The cell bodies of postganglionic parasympathetic neurones are mostly sited distant from the CNS, either in discrete ganglia located near the structures innervated, or dispersed in the walls of viscera. In the cranial part of the parasympathetic system there are four small peripheral ganglia – ciliary, pterygopalatine, submandibular and otic – that are all described on a regional basis in Section 4. These are efferent parasympathetic ganglia, unlike the trigeminal, facial, glossopharyngeal and vagal sensory ganglia, all of which are concerned exclusively with afferent impulses and contain the cell bodies of sensory neurones. The cranial parasympathetic ganglia are also traversed by afferent fibres, postganglionic sympathetic fibres and, in the case of the otic ganglion, by branchial efferent fibres; however, none of these fibres synapse in the ganglia. Postganglionic parasympathetic fibres are usually unmyeli-

nated and shorter than their counterparts in the sympathetic system because the ganglia in which the parasympathetic fibres synapse are either in or near the viscera they supply. Postganglionic parasympathetic neurones are cholinergic.

ENTERIC NERVOUS SYSTEM AND INTRINSIC NEURONES

Many peripheral autonomic ganglia contain neurones derived from the neural crest during embryonic development that are anatomically distinct from classical sympathetic and parasympathetic neurones. Connections between these intrinsic neurones allow them to sustain and modulate visceral activities by local reflex mechanisms. The enteric nervous system consists of many millions of neurones and enteric glial cells grouped into ganglionated plexuses lying in the wall of the gastrointestinal tract; ganglia containing neuronal cell bodies and glia are connected by bundles of axons to form myenteric and submucous plexuses that extend from the oesophagus to the anal sphincter. This intrinsic circuitry mediates numerous reflex functions including the contractions of the muscular coats of the gastrointestinal tract, secretion of gastric acid, intestinal transport of water and electrolytes, and the regulation of mucosal blood flow (Poole and Furness 2012). Although complex interactions occur between the enteric and sympathetic and parasympathetic nervous systems, the enteric nervous system is capable of sustaining local reflex activity independent of the CNS.

Since intrinsic neurones survive after section of the extrinsic sympathetic and parasympathetic nerves, organs that are transplanted are not truly denervated. Separation from their autonomic input often has no obvious impact on the non-striated muscle or glands innervated by autonomic fibres; contraction may be unaffected and no structural changes ensue. This has been variously attributed to the continued activity of local plexuses or to the intrinsic activity of visceral muscle. In some important instances, however, denervation does result in cessation of activity, e.g. in sweat glands, pilomotor muscle, orbital non-striated muscle and the suprarenal medulla.

VISCERAL AFFERENT PATHWAYS

Visceral afferent pathways in many ways resemble somatic afferent pathways. The peripheral processes of visceral afferents run through autonomic ganglia or plexuses, and possibly through somatic nerves. Their central processes accompany somatic afferent fibres through cranial nerves or dorsal spinal roots into the CNS, where they establish connections that mediate autonomic reflexes and visceral sensation.

General visceral afferent fibres from the viscera and blood vessels accompany their efferent counterparts, and are the peripheral processes of unipolar cell bodies located in some cranial nerve and dorsal root ganglia. They are contained in the vagus, glossopharyngeal and possibly other cranial nerves; the second to fourth sacral spinal nerves, distributed with the pelvic splanchnic nerves; and thoracic and upper lumbar spinal nerves, distributed through rami communicantes and alongside the efferent sympathetic innervation of viscera and blood vessels.

The cell bodies of vagal general visceral afferent fibres are in the superior and inferior vagal ganglia. Their peripheral processes are distributed to terminals in the pharyngeal and oesophageal walls where, acting synergistically with glossopharyngeal visceral afferents in the pharynx, they are concerned with swallowing reflexes. Vagal afferents are also believed to innervate the thyroid and parathyroid glands. In the heart, vagal afferents innervate the walls of the great vessels, the aortic bodies and pressor receptors, where they are stimulated by raised intravascular pressure. In the lungs they are distributed via the pulmonary plexuses. They supply bronchial mucosa, where they are probably involved in cough reflexes; bronchial muscle, where they encircle myocytes and end in tendrils, which are sometimes regarded as muscle spindles and which are believed to be stimulated by change in the length of myocytes; interalveolar connective tissue, where their knob-like endings, together with terminals on myocytes, may evoke Hering-Breuer reflexes; the adventitia of pulmonary arteries, where they may be pressor receptors; and the intima of pulmonary veins, where they may be chemoreceptors. Vagal visceral afferent fibres also end in the gastric and intestinal walls, digestive glands and the kidneys. Fibres ending in the gut and its ducts respond to stretch or contraction. Gastric impulses may evoke sensations of hunger and nausea.

The cell bodies of glossopharyngeal general visceral afferents are in the glossopharyngeal ganglia. Their peripheral processes innervate the

posterior lingual region, the palatine tonsils and pharynx, but they do not innervate taste buds. They also innervate the carotid sinus and the carotid body, which contain receptors sensitive to tension and changes in chemical composition of the blood. Impulses from these receptors are essential to circulatory and respiratory reflexes.

Visceral afferents that enter the spinal cord through spinal nerve roots terminate in the spinal grey matter. Almost all the central processes of vagal and glossopharyngeal afferent fibres end in the nucleus solitarius of the medulla. About 5% of the vagal afferents project directly to, and terminate in, the upper cervical spinal cord (C1–2), where they are believed to contribute to referred sensations, as well as to propriospinal mechanisms of nociceptive modulation. Visceral afferents establish connections within the CNS that mediate autonomic reflexes. In addition, afferent impulses probably mediate visceral sensations such as hunger, nausea, sexual excitement, vesical distension, etc. Visceral pain fibres may follow these routes. Although viscera are insensitive to cutting, crushing or burning, excessive tension in smooth muscle and some pathological conditions produce visceral pain. In visceral disease, vague pain may be felt near the viscus itself (visceral pain) or in a cutaneous area or other tissue whose somatic afferents enter spinal segments receiving afferents from the viscus, a phenomenon known as referred pain. If inflammation spreads from a diseased viscus to the adjacent parietal serosa (e.g. the peritoneum), somatic afferents will be stimulated, causing local somatic pain, which is commonly spasmodic. Referred pain is often associated with local cutaneous tenderness. For further reading on visceral pain, see Bielefeldt and Gebhart (2013).

Afferent fibres in pelvic splanchnic nerves innervate pelvic viscera and the distal part of the colon. Vesical receptors are widespread; those in muscle strata are associated with thickly myelinated fibres and are believed to be stretch receptors, possibly activated by contraction. Pain fibres from the bladder and proximal urethra traverse both the pelvic splanchnic nerves and the inferior hypogastric plexus, hypogastric nerves, superior hypogastric plexus and lumbar splanchnic nerves to reach their cell bodies in ganglia on the lower thoracic and upper lumbar dorsal spinal roots; the significance of this dual sensory pathway is uncertain. Lesions of the cauda equina abolish pain from vesical overdistension, but hypogastric section is ineffective. Pain fibres from the uterus traverse the hypogastric plexus and lumbar splanchnic nerves to reach somata in the lowest thoracic and upper lumbar spinal ganglia, and hypogastric division may relieve dysmenorrhoea. Afferents from the uterine cervix traverse the pelvic splanchnic nerves to reach their somata in the upper sacral spinal ganglia; stretching the cervix uteri causes pain but cauterization and biopsy excisions do not.

In general, afferent fibres that accompany pre- and postganglionic sympathetic fibres have a segmental arrangement and end in spinal cord segments from which preganglionic fibres innervate the region or viscus concerned. General visceral afferents entering thoracic and upper lumbar spinal segments are largely concerned with pain. Nociceptive impulses from the pharynx, oesophagus, stomach, intestines, kidneys, ureter, gallbladder and bile ducts seem to be carried in sympathetic pathways. Cardiac nociceptive impulses enter the spinal cord in the first to fifth thoracic spinal nerves, mainly via the middle and inferior cardiac nerves, but some fibres pass directly to the spinal nerves. It is said that there are no general visceral afferents in the superior cardiac nerves. Peripherally, the fibres pass through the cardiac plexuses and along the coronary arteries. Myocardial anoxia may evoke symptoms of angina pectoris in which pain is typically presternal, and is also referred to much of the left chest, and radiates to the left shoulder, the medial aspect of the left arm, along the left side of the neck to the jaw and occiput, and down to the epigastrium. Cardiac afferents carried in vagal cardiac branches are concerned with the reflex depression of cardiac activity. Ureteric pain fibres, also running with sympathetic fibres, are presumably involved in the agonizing renal colic that follows obstruction by calculi. Afferent fibres from the testis and ovary run through the corresponding plexuses to somata in the tenth and eleventh thoracic dorsal root ganglia.

Some primary afferent neurones, innervating the gut, lungs, heart and blood vessels, also appear to have an efferent function in that they release transmitters from their peripheral endings during the axon reflex. The primary substances released in this way are substance P, calcitonin gene-related peptide (CGRP) and ATP. These act on target cells to produce vasodilation, increased venular permeability, changes in smooth muscle contractility, degranulation of mast cells and a variety of effects on leukocytes and fibroblasts, a process collectively known as 'peripheral neurogenic inflammation'. The local release of such substances may play a trophic role in the maintenance of tissue integrity and repair in response to injury (for further reading, see Xanthos and Sandkühler (2014)).

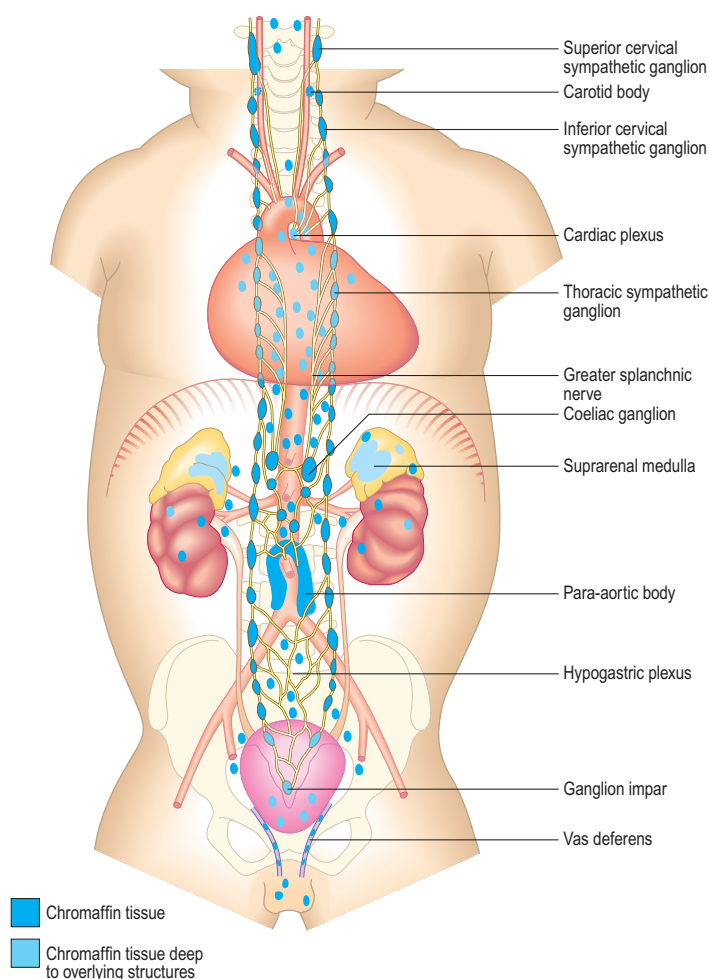


Fig. 16.14 The principal aggregations of 'classic' chromaffin tissue in the human neonate. The aggregates in pale blue lie deep to overlying structures.

PARAGANGLIA

Paraganglia are extra-suprarenal aggregations of neuroepithelial cells. They are all derived from the neural crest and all synthesize and store catecholamines (Fig. 16.14). Paraganglia can be divided into two broad categories: non-chromaffin, related to the parasympathetic system, particularly in the head and neck, where they tend to be associated with the glossopharyngeal or vagus nerves (carotid body, promontory of the middle ear, tunica adventitia of the jugular bulb, facial canal, ganglion nodosum, larynx); and chromaffin, connected with the orthosympathetic system (in the sympathetic ganglia of various viscera and a variety of thoracolumbar para-aortic retroperitoneal sites). Small paraganglia have also been reported incidentally in the urinary bladder, prostate, gallbladder, splenic capsule and mesosigmoid. Their functions differ with location: some cells act as interneurons, others as sources of neuroendocrine secretion. Paraganglionic catecholamine release occurs mainly in response to chemical rather than to neural stimuli. Extra-suprarenal chromaffin tissue is prominent in the fetus, and acts as the main source of catecholamines while the suprarenal medulla is immature. Although many paraganglia degenerate soon after birth, others persist into adulthood, often as microscopic paraganglia.

Paraganglia are well vascularized and their secretory cells are usually close to one or more fenestrated capillaries. Most have a sympathetic innervation and act as endocrine organs, e.g. suprarenal medullary chromaffin cells. Those associated with the parasympathetic system probably have activities on local nerve endings, e.g. the carotid body. Paraganglia produce regulatory peptides, particularly enkephalins, and store them as cytoplasmic granules until stimulated to release them. Their secretions may exert a local paracrine action on nearby cells, in addition to having remote endocrine effects.

Paragangliomas have been found in practically every site in which normal paraganglia are known to occur; carotid body paragangliomas are the most common and important group of extra-suprarenal paragangliomas (accounting for approximately 60% of head and neck paragangliomas). For further reading, see Rosai (2011) and Lack and Wieneke (2013).

SURFACE ANATOMY

The surface positions of cerebral structures, meningeal arteries and dural venous sinuses may be approximated via a number of standardized reference lines and planes.

Cerebral sulci and lobes

The central sulcus (Rolandic fissure) and lateral sulcus (Sylvian fissure) of the cerebral hemispheres can be landmarked via the cranium using multiple methods, the most common being that developed by Taylor and Haughton (Fig. 16.15) (Taylor and Haughton 1900, Taylor et al 1980). Recent analysis of these methods has shown them to be accurate to within a few millimetres (Reis et al 2011). Using Taylor–Haughton craniometric lines, the lateral sulcus sits on the frontozygomatic (Sylvian) line, which passes from the frontozygomatic suture to a point 75% the way along the nasion–inion midsagittal line. The lateral sulcus extends posteroinferiorly from the Sylvian point, which is located around the region of the pterion. The lateral sulcus has also been shown to align with the anterior part of the squamosal suture in a zone 2.5–4.0 cm anterior to the external acoustic meatus (Rahmah et al 2011). The position of the central sulcus is approximated via a line passing from the superior Rolandic point, situated approximately 2 cm posterior to the midpoint of the nasion–inion midsagittal line, to the inferior Rolandic point, situated at the intersection of the frontozygomatic line with the pre-auricular/condylar line. Alternatively, the position of the central sulcus can be approximated using a line passing from the superior Rolandic point to the midpoint of the zygomatic arch, with the central sulcus ending at the point of intersection of this line with the frontozygomatic line (Rhoton 2002). The superior extent of the parieto-occipital sulcus sits at the point of intersection of the frontozygomatic line and the 75% point on the nasion–inion line. The approximate positions of the lobes of the brain, and of the precentral and postcentral gyri, can be determined following localization of the central, lateral and parieto-occipital sulci (Fig. 16.16).

Vertebral column, spinal cord and meninges

The surface positions of the termination of the spinal cord and dural sac can be mapped relative to the vertebrae and their spinous processes (see Ch. 43).

Dural venous sinuses

The superior sagittal sinus passes along the nasion–inion midsagittal line on the inner surface of the cranial vault (Fig. 16.17). In the majority of subjects it deviates to the right of the sagittal plane/superior sagittal suture by up to 11 mm (Tubbs et al 2001, Samadian et al 2011). The confluence of the dural venous sinuses (torcular herophili) sits mainly to the upper right side of the inion (Sheng et al 2012). The position of the lower border of the proximal transverse sinus is best approximated via the insertion of semispinalis capitis into the cranium (Tubbs et al 2000). The course of the transverse sinus has been shown to vary widely and no single plane or surface marking can be used to predict its route accurately (Sheng et al 2012). The asterion, found at the junction of the lambdoid, occipitomastoid and parietomastoid sutures, sits over the junction between the transverse and sigmoid sinuses in 81% of subjects, inferior to it in 15% and superior to it in 4% (Sheng et al 2012).

Middle meningeal artery

The surface positions of the anterior and posterior branches of the middle meningeal artery can be determined relative to cranial landmarks (see Fig. 26.8A).



Bonus e-book images

Fig. 16.15 Cranial reference lines, craniometric points and cerebral sulci.

Fig. 16.16 Cerebral sulci, gyri and lobes.

Fig. 16.17 Cranial suture joints, fontanelles and dural venous sinuses.

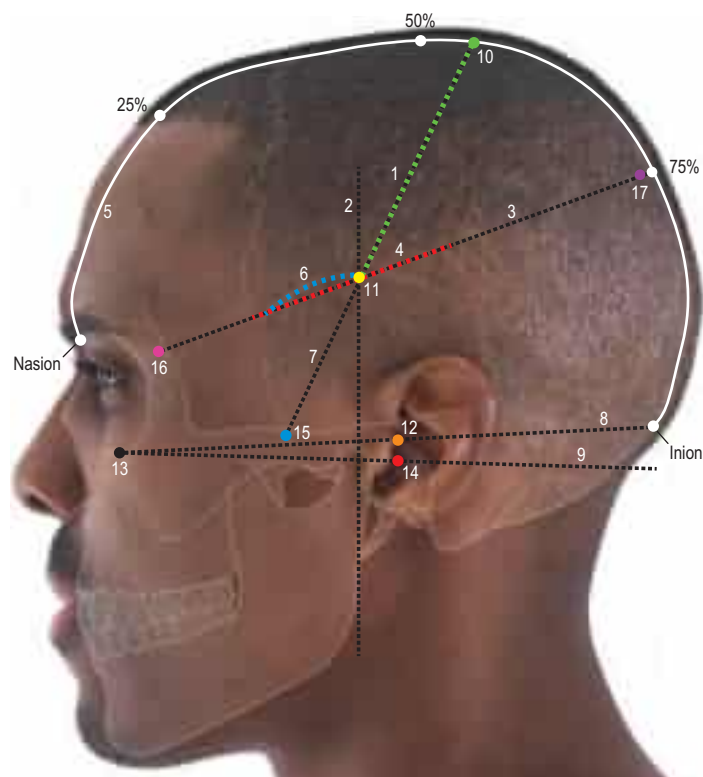


Fig. 16.15 Cranial reference lines, craniometric points and cerebral sulci. Key: 1, Rolandic line/central sulcus (green line); 2, pre-auricular/mandibular condylar line orientated perpendicular to the infra-orbitomeatal line; 3, frontozygomatic/Sylvian line passing from the frontozygomatic suture (16) to a point 75% of the way along the nasion-inion midsagittal line; 4, lateral sulcus/Sylvian fissure (red line); 5, nasion-inion midsagittal line; 6, lateral sulcus/Sylvian fissure alternative location (blue line): aligns anteriorly with the anterior part of the squamosal suture in a zone 2.5–4.0 cm anterior to the external acoustic meatus; 7, line from the superior Rolandic point (green dot) to the midpoint of the zygomatic arch (15, blue dot); 8, Reid's baseline/Frankfurt plane; 9, infra-orbitomeatal line; 10 (green dot), superior Rolandic point 2 cm posterior to the midpoint of the nasion-inion line (at the 53–55% point of the nasion-inion line); 11 (yellow dot), inferior Rolandic point; 12 (orange dot), porion; 13 (black dot), orbitale; 14 (red dot), auricular point; 15 (blue dot), midpoint of zygomatic arch; 16 (pink dot), frontozygomatic suture; 17 (purple dot), superior extent of the parieto-occipital sulcus. (Adapted with permission from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

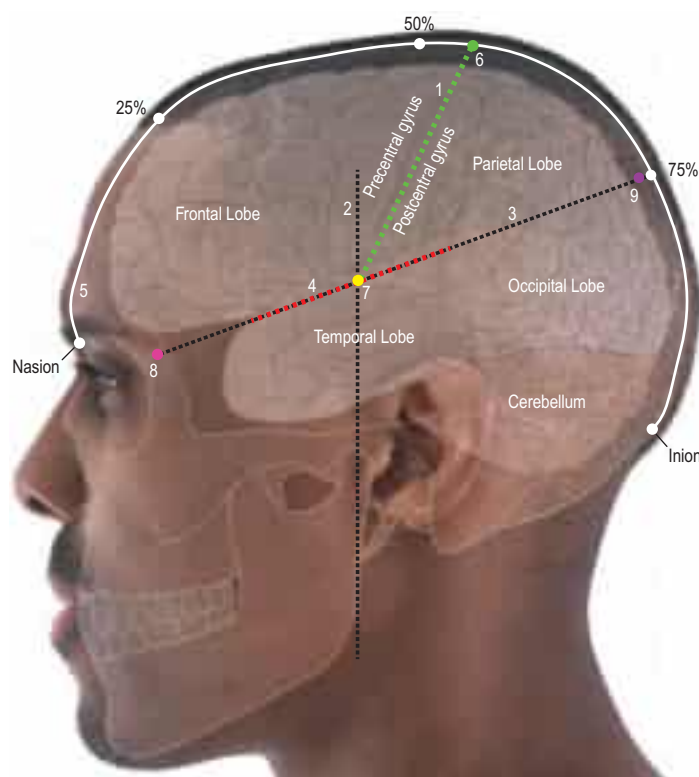


Fig. 16.16 Cerebral sulci, gyri and lobes. Key: 1, Rolandic line/central sulcus (green line); 2, pre-auricular/mandibular condylar line orientated perpendicular to the infra-orbitomeatal line; 3, frontozygomatic/Sylvian line passing from the frontozygomatic suture (8) to a point 75% of the way along the nasion-inion midsagittal line; 4, lateral sulcus/Sylvian fissure (red line); 5, nasion-inion midsagittal line; 6 (green dot), superior Rolandic point 2 cm posterior to the midpoint of the nasion-inion line (at the 53–55% point of the nasion-inion line); 7 (yellow dot), inferior Rolandic point; 8 (pink dot), frontozygomatic suture; 9 (purple dot), superior extent of the parieto-occipital sulcus. (Adapted with permission from Drake RL, Vogl AW, Mitchell A, Tibbitts R, Richardson P (eds), Gray's Atlas of Anatomy, Elsevier, Churchill Livingstone. Copyright 2008.)

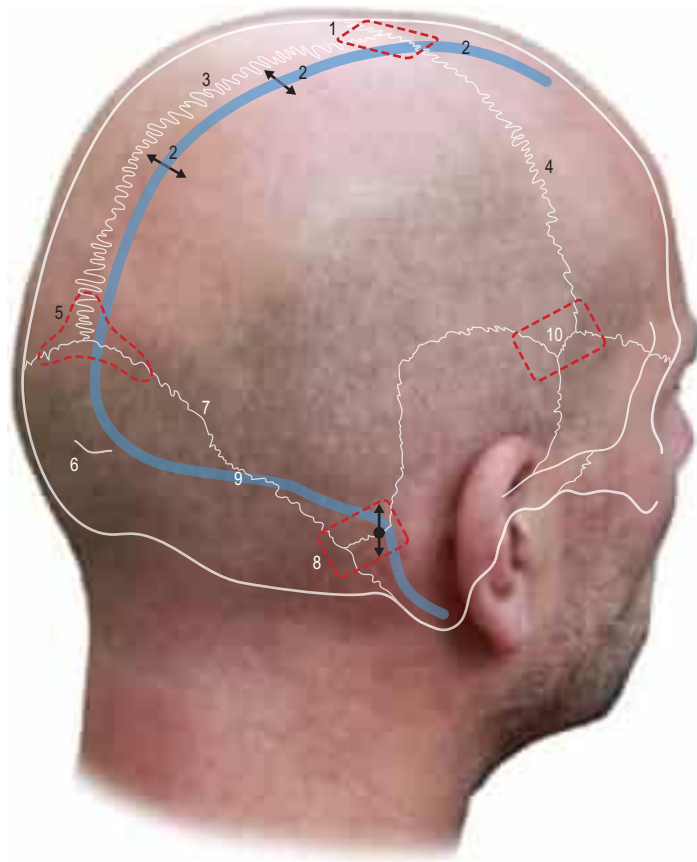


Fig. 16.17 Cranial suture joints, fontanelles and dural venous sinuses. Key: 1, bregma (site of infant anterior fontanelle); 2, superior sagittal sinus: commonly deviates to the right of the sagittal suture by up to 11 mm (black arrows); 3, sagittal suture: sits on the nasion–inion midsagittal line; 4, coronal suture; 5, lambda (site of infant posterior fontanelle); 6, inion: normally sits below and left of the confluence of sinuses and below the level of the internal occipital protuberance; 7, lambdoid suture; 8, asterion (site of infant posterolateral mastoid fontanelle): sits over (in 81%), just below (in 15%) or just above (in 4%) the transverse–sigmoid sinus junction; 9, transverse sinus: the variable course makes accurate surface localization difficult; 10, pterion (site of infant anterolateral sphenoidal fontanelle). (Adapted with permission from Tunstall R, Shah N; Surface Anatomy (Pocket Tutor series), 2012, JP Medical Ltd.)

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